



Computer Engineering Department

Course name: Web Technologies

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Topic: Report of LAB - 6

LAB 6: JavaScript: use of document.writeln(), comments, variables, local vs Global, type of operators, arithmetic, assignment, relational, logical & Bitwise Operators, Control structures, conditional statements, array, user define function, object, use if For/in

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6.1 Brief introduction of used JS::

1. JavaScript:

- JavaScript is a programming language that makes web pages interactive.
- Some main function of JS: User-side calculations, User input validation, Dynamic websites, Handling API calls, Interacting with databases, Real-time updates, Animations & UI effects.

2. var:

- Function-scoped, can be **redeclared** and **updated**.

3. let:

- Block-scoped, **cannot be redeclared** but can be **updated**.

4. const:

- Block-scoped, **cannot be redeclared or updated**.

5. comments:

- **Single-line comment** → // This is a comment.
- **Multi-line comment** → /* This is a multi-line comment */.

6. Local Variable:

- Declared inside a function/block, accessible only within it.

7. Global Variable:

- Declared outside any function, accessible everywhere.

8. Types of operators:

- **Arithmetic Operators** → + - * / % ** ++ -- .
- **Assignment Operators** → = += -= *= /= %= **=
- **Relational Operators** → == === != !== > < >= <=
- **Logical Operators** → && (AND) || (OR) ! (NOT)
- **Bitwise Operators** → & | ^ ~ << >> >>>

9. Control Structures:

- **Conditional Statements** → Execute code based on conditions (if-else, switch).
 - i.

```
switch(){
  case:
  default:
}
```
 - ii.

```
if(condition){
}
else{
}
```
- **Loops** → Repeat code execution (for, while, do-while).
 - i.

```
do{
}while(condition);
```
 - ii.

```
while(condition){
}
```
 - iii.

```
for(condition){
```

```
}
```

10. User Defined Functions:

- A user-defined function is a function created by the user to perform a specific task.

```
i. function function_name (parameters) {  
    // function body  
    return return_value;  
}
```

11. Objects:

- An object is a collection of key-value pairs where values can be of any data type.

```
i. Let obj_name = {  
    key: value;  
}
```

6.2 Exercise 1: Write a JavaScript program to use Operators

HTML:

```
<!DOCTYPE html>  
  
<html lang="en">  
  
<head>  
  
    <meta charset="UTF-8">  
  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  
    <title>Document</title>  
  
</head>  
  
<body>  
  
    <script>  
  
        let a= 2+3;  
  
        document.writeln("2+3 = ",a," ", typeof(a),"<br>");  
  
        a = 2 +"MB";  
  
        document.writeln('2 + "MB" = ',a," ", typeof(a),"<br>");
```

```

a= "hello" + "bvm";
document.writeln("hello" + "bvm" = ',a," ', typeof(a),"<br>");

a = 3 - 2;
document.writeln("3-2 = ',a," ', typeof(a),"<br>");

a = 3 - "MB";
document.writeln('3-"MB" = ',a," ', typeof(a),"<br>");

a = 3*2;
document.writeln('3*2 = ',a," ', typeof(a),"<br>");

a = 5/2;
document.writeln('5/2 = ',a," ', typeof(a),"<br>");

a = 2/0;
document.writeln('2/0 = ',a," ', typeof(a),"<br>");

a = 5%2;
document.writeln('5%2 = ',a," ', typeof(a),"<br>");

a = 2%0;
document.writeln('2%0 = ',a," ', typeof(a),"<br>");

let x = 1;
document.writeln("x = ", x," ", typeof(x),"<br>");

x++;
document.writeln("x++ = ", x," ", typeof(x),"<br>");

let y = ++x
document.writeln("y = ++x, y=", y," ", "x=",x," ", typeof(y),"<br>");

y = x++
document.writeln("y = x++, y=", y," ", "x=",x," ", typeof(y),"<br>");

x = 5;
document.writeln("x = ", x," ", typeof(x),"<br>");

x--;
document.writeln("x-- = ", x," ", typeof(x),"<br>");

y = --x
document.writeln("y = --x, y=", y," ", "x=",x," ", typeof(y),"<br>");

y = x--

```

```
document.writeln("y = x--, y=", y, " ", "x=", x, " ", typeof(y), "<br>");  
</script>  
</body>  
</html>
```

OUTPUT:

2+3 = 5 number
2 + "MB" = 2MB string
"hello" + "bvm" = hellobvm string
3-2 = 1 number
3-"MB" = NaN number
3*2 = 6 number
5/2 = 2.5 number
2/0 = Infinity number
5%2 = 1 number
2%0 = NaN number
x = 1 number
x++ =2 number
y = ++x, y=3 x=3 number
y = x++, y=3 x=4 number
x = 5 number
x-- =4 number
y = --x, y=3 x=3 number
y = x--, y=3 x=2 number

6.3 Exercise 2: Write a JavaScript program to use bitwise Operators.

HTML:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

</head>

<body>

  <script>

    let a = 5, b=2;

    document.writeln("5 | 2 = ",a|b, "<br>");

    document.writeln("5 & 2 = ",a&b, "<br>");

    document.writeln("5 ^ 2 = ",a^b, "<br>");

    document.writeln("!5 = ",!a, "<br>");

    document.writeln("!0 = ", !0 , "<br>");

    document.writeln("~0 = ", ~0 , "<br>");

    document.writeln("~5 = ", ~5 , "<br>");

    document.writeln("~2 = ", ~2 , "<br>");

    document.writeln("5 << 2 = ", a<<b , "<br>");

    document.writeln("5 >> 1 = ", 5>>1 , "<br>");

    document.writeln("1 << 5 = ", 1<<5 , "<br>");

    document.writeln("32 >> 2 = ", 32>>2 , "<br>");

  </script>

</body>

</html>
```

OUTPUT:

$5 \mid 2 = 7$

$5 \& 2 = 0$

$5 \wedge 2 = 7$

$!5 = \text{false}$

$!0 = \text{true}$

$\sim 0 = -1$

$\sim 5 = -6$

$\sim 2 = -3$

$5 \ll 2 = 20$

$5 \gg 1 = 2$

$1 \ll 5 = 32$

$32 \gg 2 = 8$

6.4 Exercise 3:Write a JavaScript program to display first N Fibonacci number .

HTML:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
</head>
```

```
<body>

<script>

    let a=0;

    let b=1;

    let n = Number(prompt("How many terms You want to print : "));

    for(let i=1;i<=n;i++){

        if(i==1){

            document.writeln("term no. ",i," = ",a,"<br>");

        }else if(i == 2){

            document.writeln("term no. ",i," = ",b,"<br>");

        }else{

            let temp = a+b;

            a = b;

            b = temp;

            document.writeln("term no. ",i," = ",b,"<br>");

        }

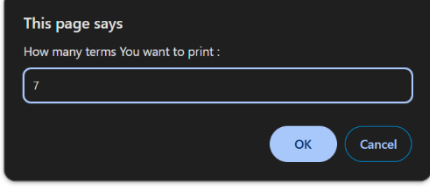
    }

</script>

</body>

</html>
```

OUTPUT:



This page says
How many terms You want to print :
7
OK Cancel

term no. 1 = 0

term no. 2 = 1

term no. 3 = 1

term no. 4 = 2

term no. 5 = 3

term no. 6 = 5

term no. 7 = 8

6.5 Exercise 4: Write JavaScript to finding all the prime number between user give ranges

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <script>

    let first = Number(prompt("Enter lower limit"));
    let last = Number(prompt("Enter Upper limit"));

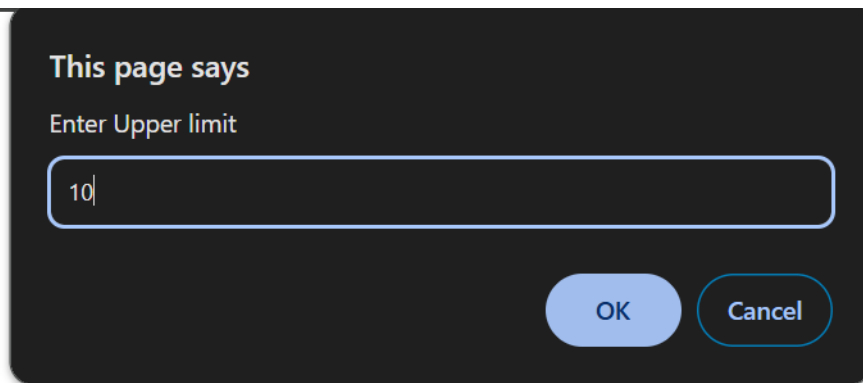
    function isPrime(num){
      if(num == 1){
        return false
      }else if(num ==2){
        return true;
      }

      for(let i=2; i<= Math.sqrt(num);i++){
        if(num % i == 0){
          return false;
        }
      }
    }
  </script>
</body>
</html>
```

```
        return true;
    }

    for(let i=first;i<=last;i++){
        if(isPrime(i)){
            document.writeln("prime = ",i,"<br>");
        }
    }
}
</script>
</body>
</html>
```

OUTPUT:



This page says

Enter Upper limit

OK Cancel

```
prime = 2
prime = 3
prime = 5
prime = 7
```

6.6 Exercise 5: Write JavaScript to use arrays methods e.g. reverse, concat, sort, join.

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <script>
    let arr = [1,2,4,2,3,5,3,1,4,3,2];
    document.writeln(" Arr = ");
    for( a of arr){
      document.write(a," ");
    }
    arr.reverse();
    document.writeln("<br>");
    document.writeln("reversed array = ");
    for( a of arr){
      document.write(a," ");
    }
    arr.sort();
    document.writeln("<br>");
    document.writeln("sorted array = ");
    for( a of arr){
      document.write(a," ");
    }
    document.writeln("<br>");
    document.writeln(arr.join(" --> "))
    document.writeln("<br>");
    document.writeln("Pop last element = ")
    arr.pop()
    document.writeln(arr);
    document.writeln("<br>");
    document.writeln("push element = ")
    arr.push(99);
    document.writeln(arr.join(" "))
  </script>
</body>
</html>
```

OUTPUT:

Arr = 1 2 4 2 3 5 3 1 4 3 2

reversed array = 2 3 4 1 3 5 3 2 4 2 1

sorted array = 1 1 2 2 2 3 3 3 4 4 5

1 --> 1 --> 2 --> 2 --> 2 --> 3 --> 3 --> 3 --> 4 --> 4 --> 5

Pop last element = 1,1,2,2,2,3,3,3,4,4

push element = 1 1 2 2 2 3 3 3 4 4 99

6.7 Exercise 6: Write JavaScript to use arrays methods e.g. reverse, concat, sort, join.

HTML:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
</head>
```

```
<body>
```

```
  <script>
```

```
    let n = Number(prompt("Enter no of elements : "));
```

```
let a = [];  
for(let i = 0;i<n;i++){  
    let num = Number(prompt("Enter element : "));  
    console.log(typeof(num))  
    a.push(num);  
}  
document.writeln("Sorted in ascending order = ");  
a.sort(Number);  
for(let i=0;i<n;i++){  
    document.writeln(a[i], " ");  
}  
document.writeln("<br>Sorted in descending order = ");  
a.reverse();  
for(let i=0;i<n;i++){  
    document.writeln(a[i], " ");  
}  
</script>  
</body>  
</html>
```

OUTPUT:

Sorted in ascending order = 1 2 3 4 5
Sorted in descending order = 5 4 3 2 1